

NAME: Sandra Sudevan
REG NO: 192424422

MODEL LAB PRACTICAL

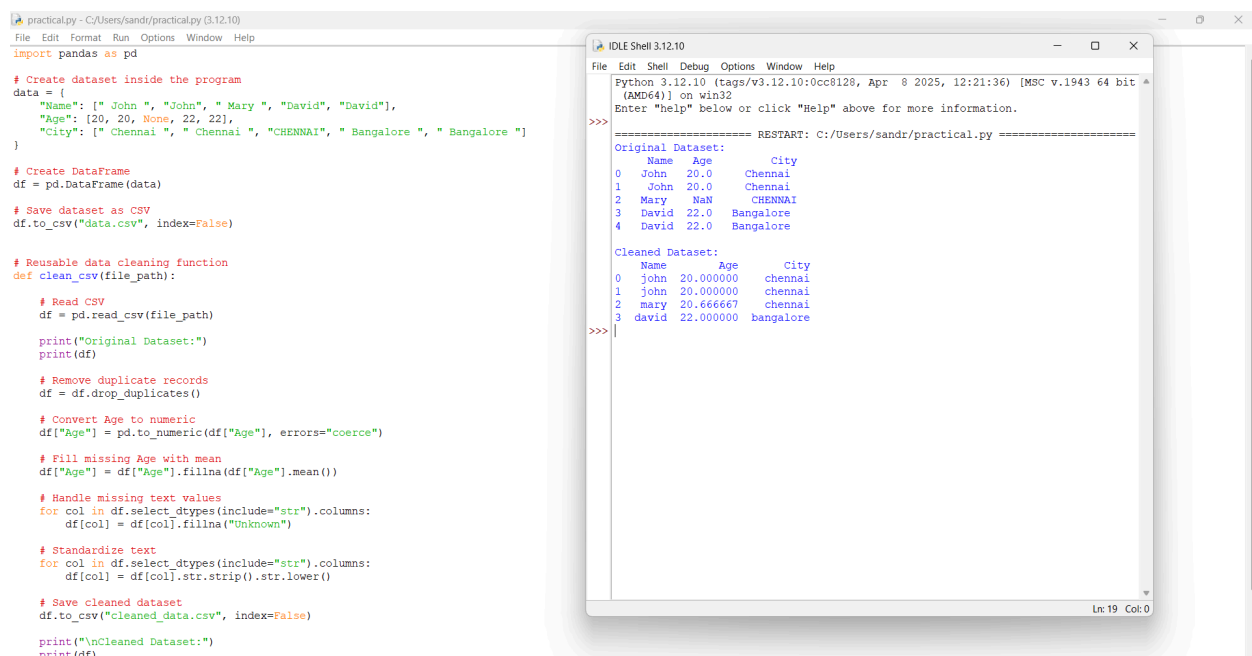
29. Automated Data Cleanup Script

AIM

To write a reusable Python function that automatically performs duplicate removal, missing-value handling, and text standardization on a CSV dataset.

PROCEDURE

1. Import the Pandas library.
2. Create a sample dataset inside the Python program.
3. Save the dataset as a CSV file.
4. Define a reusable `clean_csv()` function.
5. Read the CSV file using Pandas.
6. Remove duplicate records.
7. Handle missing values.
8. Standardize text by removing extra spaces and converting text to lowercase.
9. Display the cleaned dataset.
10. Save the cleaned dataset as a new CSV file.



The image shows a screenshot of a Python script in a text editor and its execution output in an IDE window.

Script Content:

```
practical.py - C:/Users/sandr/practical.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd

# Create dataset inside the program
data = {
    "Name": [" John ", "John", " Mary ", "David", "David"],
    "Age": [20, 20, None, 22, 22],
    "City": [" Chennai ", " Chennai ", "CHENNAI", " Bangalore ", " Bangalore "]
}

# Create DataFrame
df = pd.DataFrame(data)

# Save dataset as CSV
df.to_csv("data.csv", index=False)

# Reusable data cleaning function
def clean_csv(file_path):

    # Read CSV
    df = pd.read_csv(file_path)

    print("Original Dataset:")
    print(df)

    # Remove duplicate records
    df = df.drop_duplicates()

    # Convert Age to numeric
    df["Age"] = pd.to_numeric(df["Age"], errors="coerce")

    # Fill missing Age with mean
    df["Age"] = df["Age"].fillna(df["Age"].mean())

    # Handle missing text values
    for col in df.select_dtypes(include="str").columns:
        df[col] = df[col].fillna("Unknown")

    # Standardize text
    for col in df.select_dtypes(include="str").columns:
        df[col] = df[col].str.strip().str.lower()

    # Save cleaned dataset
    df.to_csv("Cleaned_data.csv", index=False)

    print("\nCleaned Dataset:")
    print(df)
```

Execution Output:

```
IDLE Shell 3.12.10
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/sandr/practical.py =====
Original Dataset:
   Name  Age  City
0  John  20.0  Chennai
1  John  20.0  Chennai
2  Mary   NaN  CHENNAI
3  David  22.0  Bangalore
4  David  22.0  Bangalore

Cleaned Dataset:
   Name  Age  City
0  John  20.000000  chennai
1  John  20.000000  chennai
2  mary  20.666667  chennai
3  david  22.000000  bangalore

>>>
```

RESULT

Thus, the automated data cleanup program was successfully implemented. The program removed duplicate records, filled missing values, standardized text data, and generated a cleaned CSV file successfully.